Hamstring Return-to-Play Protocol Framework

Overview: Sudden-onset hamstring injuries often occur during high-speed running or stretching (hip flexion and knee extension) but can also result from acceleration, deceleration, pressing, and indirect player contact. Injury prevention efforts should go beyond focusing solely on high-speed running, targeting the multiple activities associated with hamstring injuries.

Injury Insights (Aspetar's Surveillance, Qatar Stars League):

86% of sudden-onset hamstring injuries occur during running.

24% occur during acceleration and deceleration activities (0-10m).

Indirect contact and balance issues: Involved in 53% and 63% of injuries within the 0-10m range.

Pressing activities: Involved in 46% of injuries, with 25% occurring when the injured player pressed and 21% when they were pressed by an opponent.

- **Player contact** involved in 69% of injuries while pressing and inadequate balance involved in 82%.
- **Recommendation:** Prevention efforts should account for acceleration, deceleration, pressing, balance training, and indirect contact scenarios.

Hamstring Injury Grades and Recovery Process

Grade 1 Hamstring Injury (7-10 Days Recovery)

Initial Recovery Goals (Days 1-10):

- Focus on **pain-free, functional exercises** to restore mobility, balance, and low-impact strength.
- Begin with **low-speed, low-intensity running drills**:
    - **Speed Zone 1**: 3-9 km/hr (walking)
    - **Speed Zone 2:** 9-13 km/hr (fast walk/jog)
    - **Speed Zone 3:** 14-19 km/hr (light running)

Objective: Maintain aerobic fitness through controlled, low-speed running that limits hamstring load.

Key Components for Recovery Phase:

Acceleration/Deceleration Mechanics:

- Introduce low-intensity running drills focused on improving competency in these mechanics.

Isometric Strength Training:

- Resistance exercises: Use BOSU balls or similar equipment for core stability, to build core strength.

Proprioceptive Training:

- **Balance exercises:** Emphasize coordination and balance through proprioceptive exercises to address stability and indirect/direct-contact risks.
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Progressive Loading and Return to Participation (After 10 Days)

1. **Participation in Warm-Ups and Low-Intensity Drills:**
    - Join warm-up routines, passing, and receiving drills without intensity variations.
    - **Integrate jogging** to build volume in high-speed running distances.

2. **Progressive Speed Increase (Days 10+):**
    - Transition to Speed Zone 4 (19-24 km/hr) running to gradually load the hamstring in higher-speed running.
    - Start adding **acceleration and deceleration drills:**
        - Focus on **curved runs, short sprints, and controlled acceleration/deceleration** to target hamstrings, glutes, and lower back.
    - **Goal:** Achieve pain-free high-speed running to 80% of the player's maximum sprint speed.

3. **Return to Full Speed and Capacity (Days 15+):**

- Once able to handle 80% of max sprint speed pain-free, build capacity within Speed Zone 5 (25 km/hr and above).
- Mechanical Load: Progressively increase accelerations and decelerations to prepare for dynamic, high-stakes movements.

4. **Controlled Return to Training:**

- Reintegrate into full training, limiting exposure to **medium and large-sided games** at first.
- **Controlled Pressing:** Since pressing is a risk factor, start as a wall player in pressing transition drills, engaging only in possession to control the intensity.

5. **Long-Term Load Building (Weeks 3-4+):**

- Gradually increase play time and intensity:
    - Session durations: Begin with 15 minutes, then progress to 30, 45, and full duration as tolerated.
- Continue **balance and proprioception and strength training** throughout the return-to-play period to reinforce stability, coordination, and to consistently improve strength.

Protocol Summary:

1. **Days 1-10**: Pain-free functional exercises, low-intensity running, balance training.
2. **Days 10+:** Join warm-ups, increase speed and introduce jogging; focus on controlled, progressive load building.
3. **Days 15+:** Return to training, gradually add accelerations and decelerations, work up to full sprint capacity.
4. **Weeks 3-4+:** Controlled game integration, pressing restriction, gradual increase in session duration, and continued balance/proprioception exercises.